

ALEXANDROS MILONAKIS

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Professional Summary

Computer Science & Engineering graduate (July 2026) specializing in Machine Learning and LLM research. Combining a rigorous foundation in algorithm design and data optimization with hands-on Python expertise. Eager to apply unsupervised learning techniques and scalable software engineering practices to solve complex data challenges.

Education

University of Ioannina

Diploma in Computer Science & Engineering (Integrated Master)

Expected July 2026

Ioannina, Greece

Experience

Ontovisual

Software Engineering Intern

May 2026 – June 2026

On-Site / Ioannina, Greece

- Architected an automated typesetting engine utilizing Shape Grammars and Guillotine Partitioning algorithms to deterministically map generative AI outputs into pixel-perfect, magazine-style UI pages.
- Designed structured prompt architectures to constrain LLM outputs into strictly formatted JSON payloads, seamlessly integrating with a custom client-side validation schema.
- Engineered a fault-tolerant deserialization pipeline with dynamic overflow and look-ahead buffer logic to translate serialized AI object states into continuous, multi-page component layouts.
- Developed adaptive typography guardrails to resolve Greek text-wrapping collisions, automatically truncating text in edit mode to guarantee zero element overlaps.

PanCar Rentals

Fleet Operations & Client Service

2018 – 2022

Chania, Crete

- Utilized strategic communication to de-escalate and resolve last-minute scheduling conflicts under strict time constraints, maintaining high customer satisfaction across 5 consecutive seasons.
- Demonstrated rapid adaptability in a fast-paced operational hub by onboarding new seasonal staff and adjusting fleet logistics to meet shifting daily demands.

University of Ioannina Radio Club

IT Support & Web Administrator

2017 – 2021

Ioannina, Greece

- Fostered effective teamwork by coordinating technical setups and logistics alongside cross-functional teams to deliver successful live music festivals and university seminars.
- Maintained and updated the station's digital infrastructure and website content, ensuring high uptime and accessibility for listeners.

Technical Projects

Master's Thesis: Measuring Bias in LLM2Vec Representations

Feb. 2025 – Feb. 2026

Python, LLMs, NLP

- **Developed an automated diagnostic framework** to benchmark 6 public LLM2Vec-adapted bidirectional encoders against traditional Word2Vec and BERT baselines.
- **Engineered robust semantic subspaces** utilizing L2-normalization and PCA, demonstrating via Explained Variance Ratio that PCA provides the optimal coordinate system for mapping bias in complex hidden states.
- **Investigated model compression behaviors**, discovering that structurally pruned checkpoints (Sheared-LLaMA-1.3B) inherently “denoise” stereotypic correlations more effectively than raw parameter scaling.

Food Hazard NLP Classification (SemEval-2025) | *Python, BERT, RoBERTa, Scikit-learn*

- Developed an ensemble classification system (BERT, RoBERTa, TF-IDF weighted SVM) to predict heavily imbalanced hazard and product categories from unstructured recall reports.
- Optimized a hierarchical macro-F1 metric across 32 classes, outperforming classical ML baselines and effectively mitigating error propagation across long-tail distributions.

Technical Skills

Languages: Python, Java, C, JavaScript, HTML/CSS, SQL

Frameworks & Web: React, Redux, craft.js

AI & Data: Machine Learning, Deep Learning, NLP (LLMs), Data Mining, NumPy, Pandas, Scikit-learn

Core Concepts: Computational Geometry, Space Partitioning, Layout Algorithms, Prompt Engineering

Developer Tools: Git, VS Code, Linux, Jupyter Notebooks